

Advanced Certification in Data Science & Analytics

In Association with



Knowledge Partner



Registered TC



Guided by the industry's best minds



About Ethan's Tech

Ethans Tech is a leading IT training institute in Pune, India, dedicated to providing high-quality, industry-relevant education to individuals and Corporate aiming to excel in the IT sector. Since its inception, Ethans Tech has been committed to bridging the gap between theoretical knowledge and practical application, making it a top choice for aspiring IT professionals and Engineering college graduates. Whether you are a fresh graduate or an experienced professional looking to UpSkill, Ethans Tech offers the resources and support necessary to achieve your career goals in the dynamic IT landscape.



**Top-Rated
Institute**



**Featured for
Excellence**



**Proude Assoicate
member**



**Academic
Partner**



Ethan's Integrity

1

Awards and Recognition

Ethan's Tech: Proud Winner of the Excellence Award Winner in Python Training in 2016 & Excellence Award Winner in AWS Training in 2017 recognized by Urban Pro. Ethan's is acknowledged by Silicon India, Business Standard, Mid Day, and Outlook etc.

2

Vast Training Experience

Ethan's has been teaching advanced technology like cloud computing, analytics, and automation. Our experienced instructors, who work in the industry, have trained over 24,000 professionals and college graduates, helping them gain the skills they need for today's tech world.

3

Industry Experts Faculties

With over 56+ faculty members, each with an average of more than 10+ years of industry experience, our experts are seasoned professionals in their respective fields. 70% of our trainers hail from leading MNCs.

Pune’s 1st & Only Training Hub Powered by **3 IITs**



Ethan’s Tech proudly introduces a new era of learning as Pune’s first training institute with collaborations across three prestigious IIT-backed innovation hubs.

Our academic associations include:

- > TIH at Indian Institute Technology, Patna
- > TIH at Indian Institute Technology, Bombay
- > TIH at Indian Institute Technology, Pallakad



Technology Innovation Hub
at IIT Bombay

TIH Foundation
for IoT & IoE



Technology Innovation Hub
at IIT Patna

**Technology Innovation Hub
Vishlesan I-Hub Foundation**



Technology Innovation Hub
at IIT Palakkad

**IIT Palakkad
Technology IHub Foundation**
Driving automation for energy and safety

EDUCATE INNOVATE IMPACT

Ethans Tech x TIH - IIT Patna



Advanced Certification
From Vishlesan i-Hub, IIT Patna



Top 1% Faculty
100+ Industry-Backed Educators



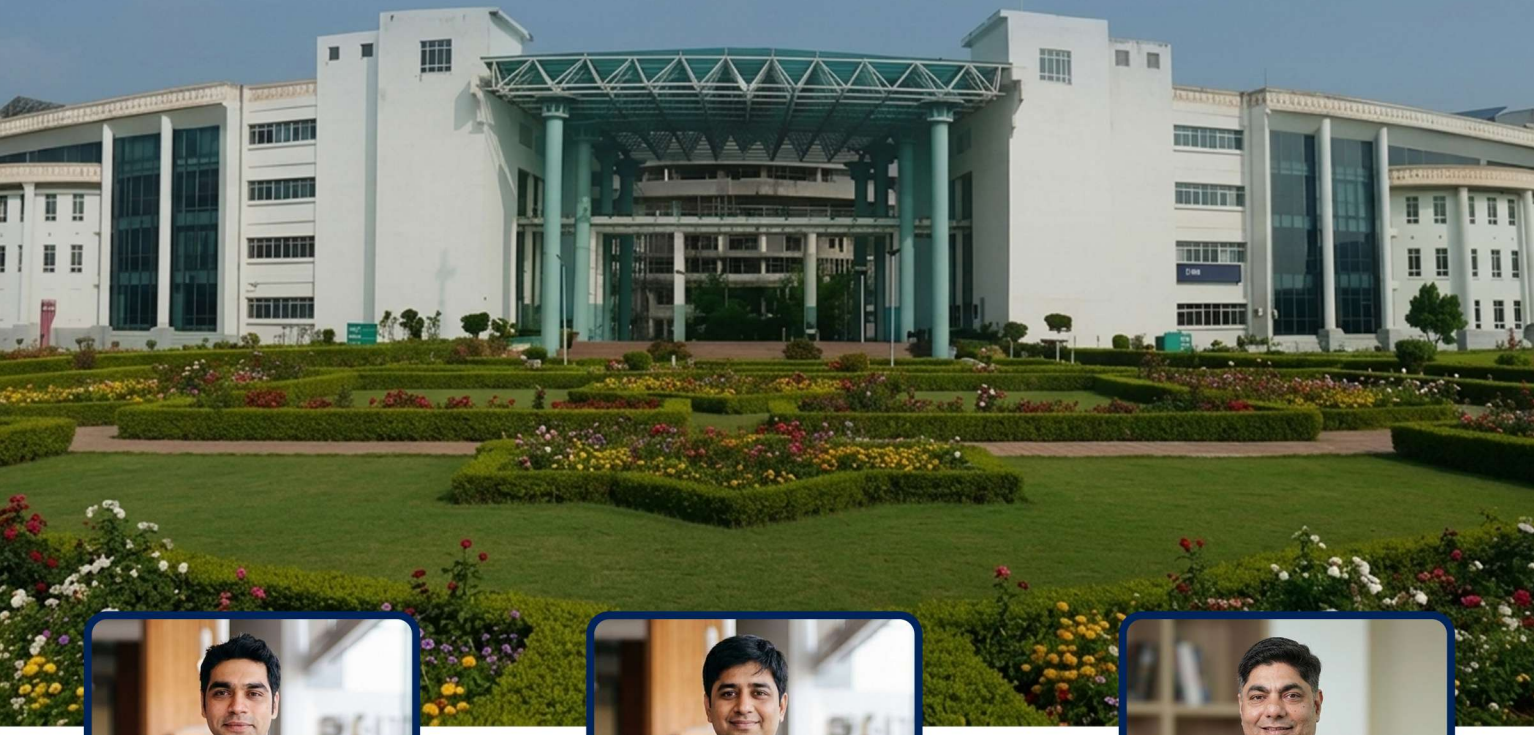
Intensive Curriculum
Industry-Relevant, Outcome-Led



Guest Lecture
By Internationally acclaimed IIT Faculties



x **TIH - IIT Patna**





Dr. Sai Kiran
CEO, i-Hub, IIT Patna



Dr. Anup Kumar Kesari
Project Director, i-Hub, IIT Patna



Mr. Jatin Miglani
CEO, Ethans Tech Pune

About FutureSkills Prime



इलेक्ट्रॉनिक्स एवं
सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF
**ELECTRONICS AND
INFORMATION TECHNOLOGY**

Why FutureSkills Prime?

 Government-recognized, industry-approved certifications

All programs are aligned with national skill frameworks, ensuring credibility and employability.

 Deep-skilling in emerging technologies

Courses are designed with hands-on learning, virtual labs, practical projects, and real-world use cases.

 Job-ready skills for modern careers

Learners gain competencies aligned with industry expectations, making them workforce-ready from day one.

 Access to incentives & career support

Eligible learners can receive government incentives, digital badges, skills passports, and career-enabling opportunities like internships, job fairs, and industry connect sessions.

FutureSkills Platform launched by Hon'ble Prime Minister Mr. Narendra Modi in Feb'18.

FutureSkills Prime (Government Initiative): Building on the initial platform, the Ministry of Electronics and IT (MeitY) and NASSCOM collaborated to launch the FutureSkills Prime program on December 24, 2019. This B2C (business-to-consumer) initiative aims to make India a digital talent nation by offering courses and certifications in 10 emerging technologies, with a government incentive program for eligible citizens.



FutureSkills – nasscom Course Alignment Certificate

Our programs are officially Course Aligned with NASSCOM FutureSkills Prime, India's leading digital skilling initiative supported by MeitY and NASSCOM. This certification ensures that the curriculum meets the National Occupational Standards (NOS) defined for emerging technology job roles.

Earning this alignment reflects that Ethan's Tech courses follow industry-approved standards, deliver job-ready skills, and are recognized across the IT-ITeS sector. It strengthens the credibility of our training and enhances the career readiness of every learner.



About the Program

Ethan's Tech Certification in Data Science & Analytics is designed to prepare you for a high-growth career in the field of Data Scientist, Data Analyst, Business Analyst, Marketing Analyst etc.

Duration:

The total duration of this program is 6 months.

Eligibility:

College students, College Graduates and ReLaunchers.
Individuals with a knack for learning and a never-give-up attitude.
No prior coding experience required

Who is it Ideal for?

Individuals looking to transition into a Data Science & Analytics.
Individuals interested in upskilling with an approved govt. certification.

Plans & Pricing

Pune's 1st & Only 4-in-1 Certification Program



~~₹72,800/-~~ ₹7925/- per month*

- Intensive Python, Data Science & Data Analytics curriculum.
- 175+ Hours of live offline classes.
- Lifetime access of all class recordings (Revised every 6 months).
- Complete LMS access with study materials and hard copy books.
- 12+ Mini Projects & 1000+ self practice quizzes, problems and assessments.
- 50+ mock interviews questions with Dedicated Placement Support.
- **Quad Certifications: i-Hub, IIT Patna, Ethan's Tech & FutureSkills Prime & NexGen Analytix Internship Certificate.**
- 63,400/- total fees, pay in 6 EMIs with 8/2 process.

* 6 EMI supported by Bajaj Finance.

Including NexGen Projects (3 months)



Program Overview

Student Workload 800+ Hours

Teaching Hours 175+ Hours

POCs & Project 12+ Projects

Total Duration 06 Months

Core Technical Modules	Duration	Learning Hours	Key Focus
Python Programming: Fundamentals to Applications	6 Weeks	45 Hours	Programming & Logic Building
Structured Query Language: Database Management & Querying	4 Weeks	25 Hours	DB Management & Queries
Advanced Data Analysis with Microsoft Excel	2 Weeks	10 Hours	Data Handling & Analysis
Microsoft Power BI: Business Intelligence & Dashboards	3 Weeks	20 Hours	Interactive Dashboards
Data Visualization with Tableau: Insights & Storytelling	2 Weeks	15 Hours	Data Visualization
Applied Statistics & Probability for Data Analysis	3 Weeks	30 Hours	Data Interpretation
Machine Learning & AI: Concepts to Practical Implementation	3 Weeks	30 Hours	Predictive Modeling
Professional English Communication Personality Development Program	8 Weeks	40 Hours	Communication & Confidence

Please note: a). English Communication & Personality development will be covered in Weekends
b). The durations of courses for individual technologies are provided as approximations and may vary from one batch to another.

Tech Covered

 **Python**
High level programming language

 **Microsoft Excel**
Basics and complex analysis

 **Microsoft Power BI**
Business intelligence tool

 **Tableau**
Data visualization tool

 **Machine Learning**
Type of Artificial Intelligence

 **SQL**
Database Programming Language

 **Artificial Intelligence**
Add Intelligence to Software

 **NumPy**
Numerical computing library

 **Pandas**
Data Analysis and Manipulations

 **Matplotlib**
Data Visualization Tool



Python Programming: Fundamentals to Applications

1. Prerequisite: Installation Process

- Python Interpreter Installation
- Python vs Anaconda Python
- IDE installation
- Introduction of Jupyter Notebook

3. Operators in Python - 1

- Arithmetic operators
- Comparison operators
- Assignment operators
- Membership and identity operators
- Data Handling in int, Str
- Indexing & Slicing in strings

4. Operators in Python - 2

- Basic operations on list
- Complex List Handling
- Basic operations on tuple
- Complex tuple Handling
- Basic operations on dict
- Complex Dict Handling

6. Conditional Statement in Python

- Introduction to Conditional Statements
- if Statement
- if-else Statement
- if-elif-else Statement
- Nested if Statements
- Comparison Operators in Conditions
- Logical Operators (and, or, not)
- Practical Examples of Conditional Statements

8. Loops in Python - for

- Introduction to for Loop
- Syntax of for Loop. Iterating Over a Sequence
- Using range() with for Loop
- Iterating Over Strings
- Iterating Over Lists, Tuples, and Sets
- Iterating Over Dictionaries
- Nested for Loops, Using break with for Loop
- Using continue with for Loop
- Using pass with for Loop
- else with for Loop

10. User Defined Function

- Introduction to User Defined Functions
- Defining a Function using def
- Calling a Function
- Function Parameters and Arguments
- Default Parameters
- Keyword Arguments
- Return Statement
- Multiple Return Values
- Scope of Variables (Local vs Global)
- lambda function

12. File Handling in Python

- Introduction to File Handling
- Opening a File (open())
- File Modes (r, w, a)
- Reading from a File (read(), readline(), readlines())
- Writing to a File (write(), writelines())
- Closing a File (close())
- Using with Statement for File Handling
- Working with Text Files

14. Object Oriented Python - 2

- Inheritance
- Method Overriding
- Polymorphism
- Special (Magic/Dunder) Methods
- Exception Handling using try and Except

16. Data Handling with Python (numpy)

- Introduction to NumPy
- Installing and Importing NumPy
- Creating NumPy Arrays
- Array Attributes (shape, size, dtype)
- Indexing and Slicing Arrays
- Array Operations and Arithmetic
- Broadcasting in NumPy
- Reshaping Arrays (reshape(), resize())
- Aggregation Functions (sum(), mean(), min(), max())
- Mathematical Functions in NumPy
- Random Number Generation
- Working with Multi-dimensional Arrays
- Saving and Loading NumPy Arrays

19. Data Analysis with Python (matplotlib)

- Introduction to Matplotlib
- Installing and Importing Matplotlib
- Understanding pyplot Module
- Creating Basic Plots
- Line Plot
- Bar Chart
- Histogram
- Scatter Plot
- Pie Chart
- Customizing Plots (Title, Labels, Legends)
- Formatting Axes and Ticks
- Multiple Plots in One Figure
- Subplots
- Grid and Style Customization

21. Data Handling with Python (JSON)(Bonus Session 2)

- Introduction to JSON Format
- Reading JSON Data
- Writing JSON Data
- Using the json Module
- Converting Python Objects to JSON (dump, dumps)
- Converting JSON to Python Objects (load, loads)

23. Data Handling with Python (SQL) (Bonus Session 4)

- Introduction to SQL with Python
- Database Concepts and Relational Databases
- Connecting Python with SQL Databases
- Using the sqlite3 Module
- Creating a Database
- Creating Tables using SQL Queries
- Inserting Data into Tables
- Reading Data from Tables (SELECT)
- Updating Records (UPDATE)
- Deleting Records (DELETE)
- Executing SQL Queries using Python
- Fetching Data (fetchone(), fetchall(), fetchmany())
- Using Python with MySQL
- Handling Database Transactions (commit(), rollback())
- Closing Database Connections

2. Introduction - Python & Its Objects

- What is Python?
- History of Python and creator Guido van Rossum
- Why Python is popular
- Applications of Python (Data Science, AI, Web, Automation)
- Python vs other languages
- Writing the first program (Hello World)
- Python syntax basics
- Comments in Python
- Indentation rules
- Variables in Python
- Naming rules
- Basic data types - int, float, str, list, dict, tuple, set, None, Bool

5. Python Built-in functions

- Introduction to Set, None and Bool Objects
- Built-in functions in Python
- input(), len(), type(), abs(), pow(), min(), max(), sum(), range(),
- enumerate(), zip(), map(), filter(), sorted(), reversed(), all(), any(), id(), help(), dir(), isinstance().

7. Loops in Python - While

- Introduction to while Loop
- Syntax of while Loop
- Flow of Execution in a while Loop
- Using Conditions in while Loop
- Infinite while Loop
- Using break with while Loop
- Using continue with while Loop
- Using pass in while Loop
- while Loop with else Statement
- Nested while Loops

9. Comprehension in Python

- Introduction to Comprehensions
- List Comprehension
- Syntax of List Comprehension
- Conditional List Comprehension
- Nested List Comprehension
- Dictionary Comprehension, Set Comprehension
- Generator Expressions

11. Python object functions

- str - lower(), upper(), strip(), replace(), split(), join(), find(), startswith(), endswith(), count(), capitalize(), title(), swapcase(), isdigit(), isalpha(), isalnum()
- list -m append(), extend(), insert(), remove(), pop(), clear(), index(), count(), sort(), reverse()
- tuple - count(), index()
- dict - keys(), values(), items(), get()

13. Object Oriented Python - 1

- Introduction to Object-Oriented Programming
- Classes and Objects
- Creating a Class & Objects
- __init__() Constructor Method
- Instance Variables and Methods
- Class Variables and Methods
- Encapsulation & Abstraction

15. Python Built in Modules

- Introduction to Python Built-in Modules
- Importing Modules (import, from, as)
- Exploring Modules using dir()
- The math Module
- The datetime Module
- The os Module
- The logging Module

17. Data Handling with Python (pandas) - 1

- Introduction to Pandas
- Installing and Importing Pandas
- Pandas Series
- Handling Pandas Series
- Pandas DataFrame
- Handling Pandas DataFrame
- Accessing Dataframes
- Reading Data from CSV, Excel, and JSON Files
- Writing Data to Files

18. Data Handling with Python (pandas) - 2

- Data Selection and Filtering
- Handling Missing Values
- Data Cleaning and Transformation
- Sorting and Ranking Data
- Grouping and Aggregation (groupby())
- Merging and Joining DataFrames
- Basic Data Analysis and Summary Statistics

20. AI API's Python (Bonus Session 1)

- Setting up Environment (Python, pip, virtual environment, API keys)
- Working with ChatGPT API (OpenAI) – basic request & response
- Gemini API Integration (Google AI) – setup and simple usage
- Prompt Engineering Basics – writing effective prompts
- Mini Project (Text summarizer or email generator)
- Error Handling & API Best Practices (rate limits, cost control)

22. Data Handling with Python (API) (Bonus Session 3)

- Introduction to APIs
- Understanding REST APIs
- HTTP Methods (GET, POST, PUT, DELETE)
- Sending API Requests using Requests
- Handling API Responses (JSON format)
- Parsing JSON Data
- Passing Parameters in API Requests
- Handling Authentication (API Keys / Tokens)
- Error Handling in API Requests
- Working with Public APIs
- Storing API Data in Python Structures
- Converting API Data to CSV / JSON Files

24. Data Handling with Python (CSV) (Bonus Session 5)

- Introduction to CSV Files
- Reading CSV Files
- Writing CSV Files
- Using the csv Module
- Working with DictReader and DictWriter
- Handling Delimiters and Headers
- Dealing with CSV Dataset



Structured Query Language: Database Management & Querying

1. Prerequisite: Introduction to MySQL & SQL

- Installation of MySQL Database
- Installation of Work Bench
- What is SQL?
- Why SQL?
- Types of Data Storage
- Files and Spreadsheets (Excel, CSV)
- Databases
- Data Warehouses
- Relational Databases
- RDBMS Core Concepts
- Tables
- Rows and Columns

3. Building Queries: Core SQL Concepts - 1

- Fundamentals of Building SQL Queries
- Session Details
- Learning Objectives
- SQL Online Compiler
- Create a Table in SQL
- Datatypes in SQL
- Numeric Types
- Character/Text Types
- Date & Time Types
- Boolean Type
- Inserting Data into the Table
- Explanation

5. Working with Multiple Tables & Relationships

- Working with Multiple Tables & Relationships
- Learning Objectives
- Create Two Tables
- Customers Table
- Orders Table
- Table Output
- Relationships Between Tables
- Primary and Foreign Keys
- Real-World Table Connections

7. Data Cleaning & Transformation with SQL

- Data Cleaning & Transformation with SQL
- Session Details
- Learning Objectives
- Understanding Raw Data & Common Data Quality Issues
- Identifying NULLs and Invalid Values
- Handling NULL Values
- Find rows with ANY NULL
- Check NULLs column-by-column
- Using IS NOT NULL
- Fill missing values using COALESCE
- NULLIF() Function in SQL Server
- When Not to Fill NULLs
- Cleaning Text Fields

9. Type Conversion & Mathematical Operations (Live Lecture, Quiz, Programming Assignment)

- SQL + Python Integration for Data Analysis
- Session Details
- Learning Objectives
- Why Should We Use SQL + Python Together
- Connecting Python to a SQL Database
- Using a Cursor in SQLite
- Running SQL Queries from Python
- Creating SQLite3 tab

11. Window Functions (Live Lecture, Quiz, Programming Assignment)

- Window Functions in SQL
- Session Details
- Learning Objectives
- What is a Window?
- Window Functions in SQL
- Breaking Down the Window Syntax
- PARTITION BY
- ORDER BY
- Aggregate Window Functions
- Ranking Functions
- Value Functions

13. AI-Powered SQL: Query Optimization & Generation (Live Lecture, Quiz, Assignment)(Bonus)

- AI-Powered SQL: Query Optimization & Generation
- Session Details
- Learning Objectives
- Why AI + SQL?
- Speed
- Accuracy
- Intelligence
- From Question to Query (NL → SQL)
- How AI Interprets a Question
- Identify Entities
- Understand Intent
- Extract Logic
- Apply SQL Patterns
- AI as Your SQL Reviewer & Optimizer

2. Introduction to SQL for Data Analytics - 1

- Primary Key
- Foreign Key
- Relationships
- Schema
- Creating a Database
- Creating a Table
- How SQL Works
- SQL Processing Flow
- SQL Commands
- SQL Command Types
- DQL (Data Query Language)
- DML (Data Manipulation Language)
- DDL (Data Definition Language)
- DCL (Data Control Language)
- TCL (Transaction Control Language)

4. Building Queries: Core SQL Concepts - 2

- Creating a Table from an Existing Table
- SQL SELECT Query
- Select Specific Columns
- Select All Columns
- SQL SELECT DISTINCT
- SQL WHERE Clause
- Operators in SQL
- Types of SQL Operators
- Comparison Operators

6. SQL Joins

- SQL Joins & Case Statements
- Session Details
- Learning Objectives
- What is a JOIN & Why Do We Need It?
- JOIN Syntax
- Types of SQL Joins
- SQL INNER JOIN
- SQL LEFT JOIN
- SQL RIGHT JOIN
- SQL FULL JOIN
- SQL SELF JOIN
- CASE Statements in SQL
- Practice Section

8. Type Conversion & Mathematical Operations

- Window Functions & CTEs in SQL
- Session Details
- Learning Objectives
- What is a Window?
- Window Functions in SQL
- Breaking Down the Window Syntax
- PARTITION BY
- ORDER BY
- Types of Window Functions in SQL
- Aggregate Window Functions
- Ranking Window Functions
- Value Functions

10. Working with Date, Time & String Functions (Live Lecture, Quiz, Programming Assignment)

- Working with Date, Time & String Functions
- Session Details
- Learning Objectives
- Understanding Real-World Challenges with Dates & Text
- SQL Date and Time Functions
- Common SQL Date & Time Data Types
- NOW()
- CURDATE()
- CURTIME()
- DATE()
- String Functions
- Pattern Matching (LIKE & Wildcards)

12. SQL + Python Integration for Data Analysis (Live Lecture, Quiz)(Bonus)

- SQL + Python Integration for Data Analysis
- Session Details
- Learning Objectives
- Why Should We Use SQL + Python Together
- Connecting Python to a SQL Database
- Using a Cursor in SQLite
- Running SQL Queries from Python
- Creating Tables using Python
- Inserting Data into Tables

14. Advanced Queries with CTEs & Pivoting (Live Lecture, Quiz)(Bonus)

- Advanced Queries with CTEs & Pivoting
- Session Details
- Learning Objectives
- What's a CTE and Why Do We Use It?
- CTEs vs Subqueries
- Chaining CTEs
- Recursive CTEs
- Pivoting Data
- Conditional Aggregation

15. Database Design & Schema Management (Live Lecture, Quiz) (Bonus)

- Database Design & Schema Management
- Session Details, Learning Objectives
- Why Database Design & Schema Thinking Matters
- What is a Database Schema?
- Types of Database Schemas, Physical Schema
- Logical Schema, View Schema
- Operational vs Analytical Databases



Advanced Data Analysis with Microsoft Excel

1. Prerequisite: Introduction to Microsoft Excel

- Excel Interface Overview
- Navigating the Ribbon
- Understanding Worksheets and Workbooks
- Creating and Saving Workbooks
- Data Entry and Editing
- Customizing the Quick Access Toolbar
- Excel File Formats
- Basic Data Types (Numbers, Text, Dates)
- Basic Excel Operations (Copy, Cut, Paste)
- Autofill and Flash Fill
- Sorting and Filtering Data
- Basic Cell Formatting (Font, Color, Alignment)
- Using the Clipboard
- Adjusting Column Width and Row Height

4. Pivot Tables & Data Analysis

- Creating Pivot Tables, Pivot Table fields & layout
- Grouping data (dates, numbers)
- Pivot Charts, Slicers & Timelines
- Calculated fields
- Basic Data Analysis tools (What-if analysis, Goal Seek), Practice exercises

6. Microsoft Excel Shortcuts (Bonus)

- Excel Shortcut Fundamentals & Navigation
- Efficient Data Entry & Editing Techniques
- Quick Formatting & Styling Shortcuts
- Advanced Navigation & Selection Methods
- Formula & Function Efficiency Shortcuts
- Data Management (Sort, Filter, Tables)
- Worksheet Handling & Organization
- Productivity Tips & Workflow Optimization
- Practical Exercises & Case Scenarios

2. Advanced Formulas & Functions

- Revision of basic formulas (SUM, IF, VLOOKUP)
- IF with multiple conditions (Nested IF, IFS)
- Logical functions (AND, OR, NOT)
- Lookup functions (VLOOKUP, HLOOKUP, XLOOKUP introduction)
- INDEX & MATCH (better alternative to VLOOKUP)
- Text functions (LEFT, RIGHT, MID, LEN, CONCAT, TRIM)

3. Data Management & Data Cleaning

- Sorting & Filtering (Advanced Filter)
- Data Validation (drop-downs, restrictions)
- Remove Duplicates
- Text to Columns
- Flash Fill
- Conditional Formatting (rules, color scales, icons)
- Working with Tables (structured references)
- Practice exercises

5. Advanced Tools & Automation

- Introduction to Macros (recording & running)
- Absolute vs Relative referencing in macros
- Basic VBA understanding (optional intro)
- Dashboard creation basics
- Charts (advanced charts, combo charts)
- Protecting sheets/workbooks
- Tips, shortcuts & real-world case study
- Final practice / mini project



Microsoft Power BI: Business Intelligence & Dashboards

1. Power BI Overview & Data Connection

- Introduction to Power BI (Desktop, Service, Mobile)
- Data sources (Excel, CSV, SQL, Web)
- Import vs DirectQuery
- Power BI interface overview
- Loading and previewing data
- Basic data model introduction
- Practice exercise

3. Data Modeling

- Relationships (one-to-many, many-to-many)
- Cardinality & cross filter direction
- Star schema vs snowflake schema
- Creating & managing relationships
- Date table creation
- Hierarchies
- Practice exercise

5. Advanced DAX

- Time intelligence functions (YTD, MTD, SAMEPERIODLASTYEAR)
- ALL, ALLEXCEPT
- Variables in DAX
- Conditional logic (IF, SWITCH)
- Ranking functions
- Performance basics
- Practice exercise

7. Advanced Features & Power BI Service

- Publishing reports to Power BI Service
- Workspaces & sharing
- Row-Level Security (RLS)
- Gateways (intro)
- Scheduled refresh
- Bookmarks & buttons
- App creation
- Practice exercise

9. PowerBI Services (Bonus)

- Overview of Power BI Service & Cloud Environment
- Report Publishing & Dataset Management
- Dashboard Creation & Customization
- Data Refresh & Automation Setup
- Collaboration & Report Sharing
- Workspace & Access Control Management
- Integration with Cloud Data Sources
- App Creation & Distribution
- Data Security & Governance Basics
- Practical Scenarios & Case Studies

2. Power Query (Data Transformation)

- Power Query Editor overview
- Data cleaning (remove rows/columns, rename, change data types)
- Handling missing values
- Splitting & merging columns
- Pivot & Unpivot
- Adding custom columns
- Applied steps & query dependencies
- Practice exercise

4. DAX Fundamentals

- Introduction to DAX
- Calculated columns vs measures
- Basic DAX functions (SUM, COUNT, AVERAGE)
- CALCULATE function
- FILTER function
- Aggregations & simple KPIs
- Practice exercise

6. Data Visualization & Dashboard Design

- Visual types (bar, line, pie, matrix, cards)
- Formatting visuals
- Drill-down & drill-through
- Tooltips
- Slicers & filters
- Page navigation
- Design best practices (UX/UI)
- Practice exercise

8. Real-World Project & Optimization

- End-to-end dashboard project
- Data cleaning + modeling + DAX + visuals
- Performance optimization tips
- Best practices for reports
- Common interview questions
- Troubleshooting scenarios
- Final project presentation



Data Visualization - Tableau: Insights & Storytelling

1. Tableau Fundamentals & Data Connection

- Introduction to Tableau ecosystem (Desktop, Server, Public)
- Connecting to data sources (Excel, CSV, SQL)
- Live vs Extract
- Data source page overview
- Dimensions vs Measures
- Basic visualizations (bar, line, pie)
- Show Me panel
- Practice exercise

4. Calculations (Intermediate)

- Calculated fields deep dive
- Aggregate vs non-aggregate functions
- String, date, and logical functions
- Table calculations (running total, percent of total)
- Quick table calculations
- Practice exercise

5. Advanced Calculations

- Level of Detail (LOD) expressions (FIXED, INCLUDE, EXCLUDE)
- Nested calculations
- Conditional calculations
- Ranking & window functions
- Cohort analysis basics
- Practice exercise

2. Data Preparation & Cleaning

- Data Interpreter
- Sorting & filtering
- Splitting & grouping data
- Pivoting data
- Handling null values
- Creating calculated fields (basic)
- Aliases & default properties
- Practice exercise

3. Data Modeling & Joins

- Joins (inner, left, right, full)
- Union, Relationships vs joins, Blending data
- Cross-database joins
- Data source filters, Extract vs live performance considerations
- Practice exercise

6. Advanced Visualizations

- Dual axis charts, Combo charts
- Maps (symbol, filled maps)
- Highlight tables & heatmaps
- Parameters (dynamic contro)

7. Tableau Server(Bonus)

- Introduction to Tableau Server (platform overview)
- Publishing Workbooks & Data Sources
- Creating & Managing Projects
- User Roles & Permissions Management
- Sharing Dashboards & Collaboration
- Scheduling Data Refresh & Extracts
- Managing Sites & Content Organization
- Tableau Server Interface & Navigation
- Data Security & Governance Basics
- Real-world Use Cases & Practice



Applied Statistics & Probability for Data Analysis

1. Introduction to Statistics & Data

- What is Statistics? Why it matters in Data Science
- Types of Statistics: Descriptive vs Inferential
- Types of Data: Qualitative vs Quantitative
- Levels of Measurement: Nominal, Ordinal, Interval, Ratio
- Population vs Sample — key differences
- Variables: Independent, Dependent, Confounding
- Overview of tools: Python (NumPy, Pandas, SciPy)

3. Descriptive Statistics - Measures of Spread & Shape

- Range and its limitations
- Variance and Standard Deviation: Formula and intuition
- Interquartile Range (IQR) and the Five-Number Summary
- Coefficient of Variation
- Skewness: Left-skewed vs Right-skewed distributions
- Kurtosis: Understanding the tail behavior
- Computing spread and shape metrics using Python

5. Introduction to Probability

- What is Probability? Intuition and formal definition
- Basic Terminology: Experiment, Outcome, Event, Sample Space
- Classical vs Empirical vs Subjective Probability
- Complement Rule
- Addition Rule (mutually exclusive and non-exclusive events)
- Multiplication Rule (independent and dependent events)
- Practical examples with coin, dice and cards

7. Probability Distributions - Discrete

- What is a Probability Distribution? PMF and CDF
- Bernoulli Distribution
- Binomial Distribution: Formula, parameters and examples
- Poisson Distribution: Counting events over time/space
- Expected Value and Variance of discrete distributions
- Plotting discrete distributions using Python (SciPy, Matplotlib)
- Practical examples: predicting counts and binary outcomes

10. Confidence Intervals & Estimation

- Point Estimate vs Interval Estimate
- What is a Confidence Interval? Interpreting 95% CI correctly
- Margin of Error
- Z-interval: CI when population std is known
- t-interval: CI when population std is unknown
- Effect of sample size on confidence interval width
- Calculating Confidence Intervals in Python (SciPy)

11. Hypothesis Testing & Statistical Tests

- What is Hypothesis Testing? Steps and framework
- Null Hypothesis (H0) and Alternate Hypothesis (H1)
- Type I Error (False Positive) and Type II Error (False Negative)
- p-value, Significance Level (alpha) and decision rules
- One-tailed vs Two-tailed Tests
- Z-test, One-sample t-test, Independent t-test, Paired t-test
- Chi-Square Test for categorical data independence
- Performing all tests in Python (SciPy.stats) with real examples

2. Descriptive Statistics - Measures of Central Tendency

- Mean: Definition, formula and calculation
- Median: Definition, formula and when to use it
- Mode: Definition and use cases
- Weighted Mean
- When to use Mean vs Median vs Mode
- Effect of outliers on central tendency
- Calculating Mean, Median, Mode using Python (NumPy & SciPy)

4. Data Visualization for Statistics

- Why visualize data before analyzing?
- Histogram: Understanding data distribution
- Box Plot: Visualizing spread, IQR and outliers
- Bar Chart and Pie Chart for categorical data
- Scatter Plot: Spotting relationships between variables
- Heatmaps for correlation overview
- Using Matplotlib and Seaborn — hands-on with real dataset

6. Conditional Probability & Bayes' Theorem

- Conditional Probability: $P(A|B)$ — definition and formula
- Independent vs Dependent Events
- Law of Total Probability
- Bayes' Theorem: Formula and step-by-step derivation
- Intuitive understanding with a medical test example
- DS use cases: Spam Filters, Naive Bayes Classifier
- Solving Bayes' Theorem problems in Python

8. Probability Distributions - Continuous

- PDF vs PMF — key difference
- Uniform Distribution
- Normal (Gaussian) Distribution: Shape, parameters and properties
- The 68-95-99.7 Rule (Empirical Rule)
- Standard Normal Distribution and Z-scores
- Exponential Distribution and its DS applications
- Plotting and working with continuous distributions in Python

9. Sampling & Central Limit Theorem

- What is sampling? Why not use the whole population?
- Random, Stratified, Systematic and Cluster Sampling
- Sampling Error and Bias
- Sampling Distribution of the Mean
- Central Limit Theorem (CLT): Statement and intuition
- Why CLT is the backbone of inferential statistics
- Simulating CLT using Python (NumPy)

12. Correlation & Capstone Practice

- Covariance: Definition, formula and interpretation
- Pearson Correlation Coefficient
- Spearman Rank Correlation (for non-linear relationships)
- Correlation vs Causation — a critical distinction for DS
- Visualizing correlation matrices using Seaborn Heatmaps
- End-to-end capstone: EDA + descriptive stats + distributions + hypothesis testing + correlation
- Communicating statistical findings clearly to non-technical audiences



Machine Learning & AI: Concepts to Practical Implements

1. Introduction to Machine Learning

- What is ML? Difference between AI, ML and Deep Learning
- Types of ML: Supervised, Unsupervised
- ML Workflow: Data → Train → Evaluate → Predict
- Key terms: Features, Labels, Training set, Test set
- Overfitting vs Underfitting — visual intuition

3. Regression - Linear & Multiple

- What is Regression? Predicting continuous values
- Simple Linear Regression: equation, slope and intercept
- Multiple Linear Regression: using multiple features
- Evaluation metrics: MAE, RMSE, R-squared
- Hands-on: Train and evaluate a Linear Regression model

5. Classification - Logistic Regression

- What is Classification? Binary vs Multi-class problems
- Logistic Regression: Sigmoid function and decision boundary
- Confusion Matrix: TP, TN, FP, FN explained
- Evaluation metrics: Accuracy, Precision, Recall, F1-Score
- Hands-on: Binary and multi-class classification with Logistic Regression

7. Classification - SVM

- Support Vector Machine: hyperplane and margin concept
- Hard Margin vs Soft Margin SVM
- Kernel Trick: Linear, RBF and Polynomial kernels (intuition)
- SVM for multi-class classification
- Hands-on: SVM with different kernels — compare decision boundaries

9. Clustering - K-Means

- What is Clustering? Unsupervised learning use cases
- K-Means algorithm: initialise → assign → update → repeat
- Choosing K: Elbow Method and Inertia
- Limitations of K-Means: outliers and non-spherical clusters
- Hands-on: K-Means clustering with scatter plot visualisation

11. Dimensionality Reduction - PCA

- Curse of Dimensionality: why too many features hurt models
- Principal Component Analysis (PCA): intuition and purpose
- Explained Variance Ratio: choosing number of components
- Applying PCA before classification and clustering
- Hands-on: PCA on a high-dimensional dataset — visualise in 2D

2. Data Preprocessing & Feature Engineering

- Handling missing values: drop vs mean/median/mode imputation
- Encoding categoricals: Label Encoding and One-Hot Encoding
- Feature Scaling: MinMax Normalization vs Standardization
- Train-Test Split and why shuffling matters
- Hands-on: Build a full preprocessing pipeline in scikit-learn

4. Regression - Polynomial & Regularization

- Polynomial Regression: fitting curves to non-linear data
- Problem of overfitting with high-degree polynomials
- Ridge Regression (L2): shrinking large coefficients
- Lasso Regression (L1): coefficient shrinkage + feature selection
- Hands-on: Compare Linear vs Ridge vs Lasso on the same dataset

6. Classification - KNN & Naive Bayes

- K-Nearest Neighbours (KNN): distance-based prediction intuition
- Choosing the right K with cross-validation
- Naive Bayes: applying Bayes' theorem to classification
- Gaussian vs Multinomial Naive Bayes — when to use each
- Hands-on: KNN vs Naive Bayes comparison on same dataset

8. Classification - Decision Trees & Random Forest

- Decision Tree: splitting criteria — Gini Impurity and Information Gain
- Visualising a trained Decision Tree
- Overfitting in trees and pruning concept
- Random Forest: ensemble of trees and feature importance
- Hands-on: Decision Tree vs Random Forest — accuracy and feature importance

10. Clustering - Hierarchical & DBSCAN

- Hierarchical Clustering: Agglomerative approach step-by-step
- Dendrogram: reading and cutting to select cluster count
- DBSCAN: Core, Border and Noise points concept
- When to use DBSCAN vs K-Means vs Hierarchical
- Hands-on: Compare all three clustering methods on one dataset

12. Model Evaluation & Hyperparameter Tuning

- K-Fold and Stratified K-Fold Cross Validation
- ROC Curve and AUC Score for classifier comparison
- Handling class imbalance: class_weight parameter
- Hyperparameter tuning with GridSearchCV
- Hands-on: Full model selection pipeline — tune, validate and compare

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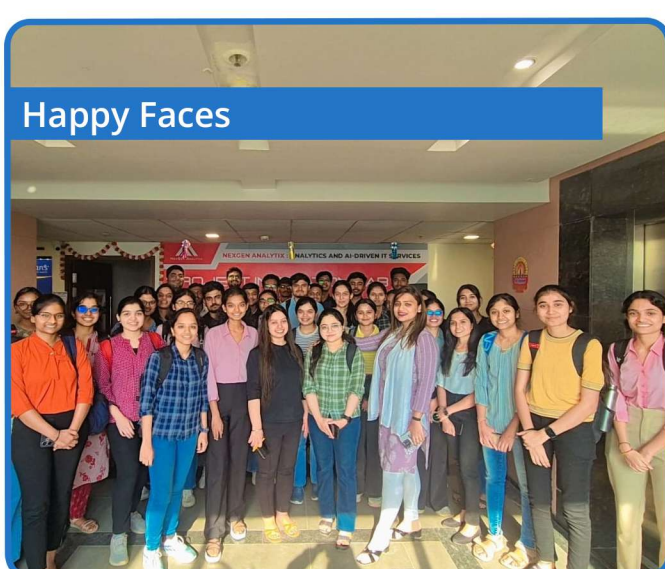
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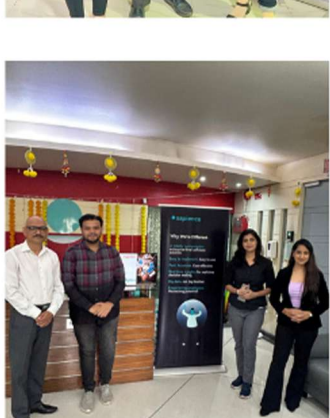
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